Nam 2019 myeloproliferative neoplasms — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates all 10 samples + both group leaves: per-neighbourhood iLISI 1.43→3.90 of a 4.94 perfect-mixing ceiling. One connected component (1 speck removed). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

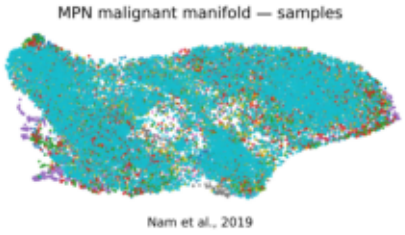
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.23), both oppose Crest (Gast↔Crest r=−0.73, pSom↔Crest r=−0.71). Along pseudotime Gast/pSom fall (ρ −0.63/−0.66), Crest rises to a terminal tip (decile pt=0.86). Inverse terminal gradient, no Gast-arm inversion.

Gate 3 — cross-repression → NA

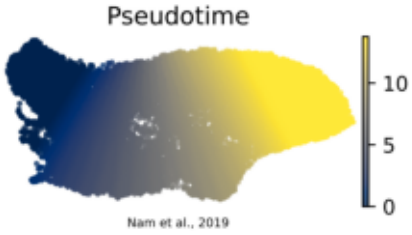
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks across Nodes=15/20/30). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

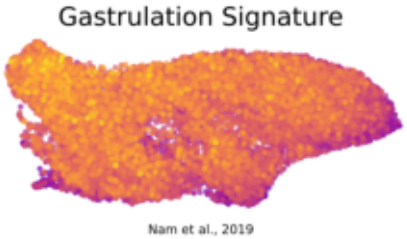
Manifold — samples



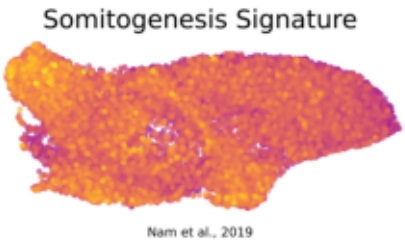
Pseudotime



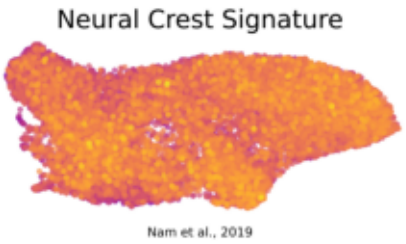
Gastrulation



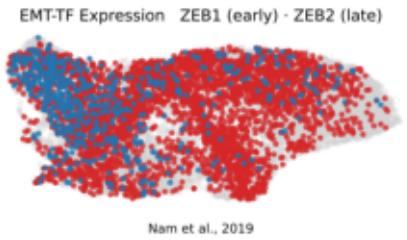
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Myeloproliferative neoplasms (CALR-mut ET/MF) (Nam et al., 2019), 10x — 22,605 malignant cells over 10 samples (2 group leaves); low-cycling Gast/pSom root pole and a cycling Crest terminus, one continuous manifold. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB1 (early/root) → ZEB2 (terminus).

Parameter choices

Sample threshold	≥100 cells (no cliff) → 10/10 samples, 22,606 cells	**Overdispersed genes**	2,162 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (iLISI plateau 5–10, declines at 20)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 1 island removed → 22,605
Geometry	non-branching curve (N=20, λ=0.3, μ=100): 2 tips / 0 forks; corr(t,DC1)=0.98	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.036 · pSom +0.008 · Crest +0.009 (Linear-pSom/Crest)	**EMT-TFs**	ZEB1 (early/root) → ZEB2 (late/terminus); detection-gated, FOXC/PRRX/SNAI in panel